

Detailed Cryptanalysis and Decryption Walkthrough

Executive Summary

The provided ciphertext appeared to be encrypted using more than one classical cryptographic technique. Analysis of the character frequencies suggested that the underlying plaintext was English, but direct frequency analysis did not immediately reveal meaningful words. Further investigation showed that the ciphertext had been processed through a two-stage encryption mechanism consisting of a block transposition (permutation) cipher followed by a monoalphabetic substitution cipher. By reversing these operations in the correct order, the original message was successfully recovered.

1. Ciphertext Provided

QMNJVSA NV WEWC FLCT VPRJ TJ TVVPLVL FV XJA VQILDHC XMLNVC NACYCLPA FC
GYT VFWV FV WGQYP PQQ PQCS Y WSQ RX QMNJVAFY CGV TLVHF CW TYL AEUQ FV
XJA TKBV CQNSQS LHF AVAWNC CV EAS FUQB QVQ TC YLLRQR XXWA CFY PSDC UQF
AVRQC GEFQ PYAT TRAC XWV TAA WWD DV EAS FLCBQ VD TRAWM VUPQ QUW X
DECGQCWT YQ YAFL VLQS YQKLHQ SNAFQ VML LHVQPAWR NQG VFUSR EC WAWY QP
FN WGAFDWG

2. Identification of Encryption Scheme

Inspection of the ciphertext revealed that letter frequencies resembled natural English, indicating that the text was not produced by a modern encryption algorithm. The challenge statement also hinted at a two-layer encryption scheme. The supplied source code confirmed that the ciphertext was first subjected to a block permutation cipher and then transformed using a substitution cipher.

Therefore, the encryption system consists of the following layers:

- 1 Layer 1: Block Permutation / Transposition Cipher (block size = 5)
- 2 Layer 2: Monoalphabetic Substitution Cipher

3. Encryption Keys Recovered

Permutation Key

Block Size = 5

Permutation Order = [3, 2, 4, 0, 1]

Substitution Key

a→q, b→j, c→e, d→p, e→v, f→s, g→g, h→f, i→c, j→k, k→m, l→t, m→u, n→y, o→w, p→h, q→i,
r→n, s→l, t→a, u→d, v→b, w→r, x→o, y→x, z→z

4. Decryption Methodology

- 1 Remove all spaces from the ciphertext and create a continuous stream of characters.
- 2 Divide the resulting text into blocks of five characters.
- 3 Apply the inverse block permutation using the permutation order [3,2,4,0,1].
- 4 Construct the inverse substitution table from the provided substitution key.
- 5 Replace every character using the inverse substitution mapping.

- 6 Convert the recovered text to uppercase and restore word boundaries.
- 7 Validate the result by checking whether the recovered text forms meaningful English sentences.

5. Code Logic Analysis

The function `decrypt_permutation(ciphertext, order)` processes the ciphertext in blocks of five characters. For each block, characters are rearranged according to the supplied order. After permutation reversal, an inverse substitution dictionary is generated automatically using the expression `{v:k for k,v in key.items()}`. This inverse mapping converts ciphertext letters back to their original plaintext equivalents. Applying this inverse mapping across the entire permuted text reveals the final message.

6. Recovered Plaintext

BREAKER OF THIS CODE WILL BE BLESSED BY THE SQUEAKY SPIRIT RESIDING IN THE HOLE GO AHEAD AND FIND A WAY OF BREAKING THE SPELL ON HIM CAST BY THE EVIL JAFFAR THE SPIRIT OF THE CAVE MAN IS ALWAYS WITH YOU FIND THE MAGIC WAND THAT WILL LET YOU OUT OF THE CAVES IT WOULD MAKE YOU A MAGICIAN NO LESS THAN JAFFAR SPEAK THE PASSWORD THE MAGIC OF WAND TO GO THROUGH

7. Final Answers

(i) Recovered Plaintext:

BREAKER OF THIS CODE WILL BE BLESSED BY THE SQUEAKY SPIRIT RESIDING IN THE HOLE GO AHEAD AND FIND A WAY OF BREAKING THE SPELL ON HIM CAST BY THE EVIL JAFFAR THE SPIRIT OF THE CAVE MAN IS ALWAYS WITH YOU FIND THE MAGIC WAND THAT WILL LET YOU OUT OF THE CAVES IT WOULD MAKE YOU A MAGICIAN NO LESS THAN JAFFAR SPEAK THE PASSWORD THE MAGIC OF WAND TO GO THROUGH

(ii) Encryption Algorithm/Cipher:

Two-layer Classical Cipher (Block Permutation Cipher + Monoalphabetic Substitution Cipher)

(iii) Encryption Keys:

Permutation Order = [3,2,4,0,1] and the substitution mapping listed above.

(iv) Solution Approach:

Remove spaces → Reverse permutation → Apply inverse substitution → Recover plaintext.